



ANALYZING GLOBAL TECHNOLOGY TRENDS: INSIGHTS FROM THE STACK OVERFLOW DEVELOPER SURVEY

IBM Data Analyst Capstone Project

Associate Data Analyst – Technology Consulting

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ANKIT SAMANT

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EXECUTIVE SUMMARY

- Objective: Extract insights from Stack Overflow Developer Survey to inform tech strategy.
- Approach
 - Python (Pandas, NumPy, Matplotlib), SQL, Jupyter
 - Dashboards in IBM Cognos / Looker Studio
- Key Findings
 - Most-used: Python, JavaScript, SQL
 - Fast-rising interest: Go, TypeScript, Rust
 - DB leaders: MySQL, PostgreSQL; MongoDB & cloud gaining
- Outcome: Actionable trends to guide hiring, training, and tech investments.

INTRODUCTION

- Why this matters: Tech stacks evolve quickly; decisions must be data-driven.
- Scope
 - Current usage vs. future learning intent
 - Languages, databases, platforms, frameworks
 - Global developer demographics
- Deliverables: Clean dataset, EDA, dashboards, and presentation of insights.

LITERATURE REVIEW

- Industry themes

- AI/automation, cloud-native architectures, data literacy

- Survey value

- Stack Overflow data tracks annual adoption & interest patterns

- Gap addressed

- Interactive dashboards & concise visuals to aid decision-makers

METHODOLOGY (OVERVIEW)

- Data Collection

- Stack Overflow survey dataset; API/web scraping for validation

- Tools

- Python, SQL, Jupyter
 - IBM Cognos / Looker Studio
 - Matplotlib for visuals

METHODOLOGY – DATA CLEANING & PREPARATION

● Challenges

- Extensive missing values in CompTotal & ConvertedComp
- YearsCodePro and Age stored as text
- EmploymentType with conflicting/multiple entries

● Fixes

- Type conversions; parsing multi-select fields
- Imputation or row filtering depending on analysis
- Normalization & deduplication

METHODOLOGY – EDA NOTES

- EDA tasks

- Descriptive stats; distributions; correlations

- Plotting constraints

- Null-heavy fields skew histograms/scatter plots
- Imputation (mean/median) vs. dropping rows can change shapes & trends

- Careful documentation of assumptions to preserve interpretability.

EDA RESULTS – PROGRAMMING LANGUAGES

- Current usage

- Leaders: Python, JavaScript, SQL

- Future interest

- Rising: Go, TypeScript, Rust

- Implication: Balance upskilling in dominant & emerging stacks.

EDA RESULTS – DATABASES & PLATFORMS

- Databases

- Leaders: MySQL, PostgreSQL
- Momentum: MongoDB, cloud databases

- Platforms

- Trend toward cloud-native and distributed systems

- Implication: Invest in scalable data infrastructure.

EDA RESULTS – FRAMEWORKS & TOOLS

- Front-end

- React, Angular remain popular

- Back-end & Data

- Node.js, Django; ML tools interest (TensorFlow, PyTorch)

- Implication: Support cross-functional tooling for web + data/ML.

EDA RESULTS – DEMOGRAPHICS

- Age & Experience

- Majority 25–34 years; 5–10 years experience

- Employment Type

- Full-time, freelance, and student roles

- Implication: Tailor learning programs and hiring to profiles.

DASHBOARDS – OVERVIEW

- Current Technology Usage

- Top languages, DBs, frameworks, platforms

- Future Technology Trends

- Intent to learn; momentum indicators

- Demographics

- Age, YearsCodePro, EmploymentType

- Exported as PDFs for sharing & portfolio.

DASHBOARDS – DESIGN CHOICES

- Clarity

- Titles, labels, legends; minimal clutter

- Interactivity

- Filters for demographics/regions/roles

- Consistency

- Aligned color families; uniform scales

- Data Quality

- Preprocessing to avoid misrepresenting trends

DISCUSSION OF INSIGHTS (CONDENSED)

● Body

- Python, JavaScript, SQL dominate; Go/TypeScript/Rust rising
- MySQL/PostgreSQL lead; MongoDB & cloud DBs gaining
- Young, skilled workforce (25–34; 5–10 yrs exp)

● Data caveats

- Nulls in CompTotal/ConvertedComp/YearsCodePro; EmploymentType inconsistencies
- Cleaning choices can shift visual trends—document assumptions

CONCLUSION & RECOMMENDATIONS

● Conclusion

- Clear adoption leaders and identifiable emerging tech momentum
- Dashboards enable rapid, stakeholder-friendly insights

● Recommendations

- Upskill in dominant & emerging languages
- Adopt cloud/NoSQL where appropriate
- Use dashboards for ongoing tracking
- Align hiring & L&D to demographics and trend signals

APPENDIX – DATA ISSUES & ASSUMPTIONS

● Key issues

- Missing: CompTotal, ConvertedComp
- Text types: Age, YearsCodePro
- Multi-select & conflicting EmploymentType entries

● Assumptions

- Selective imputation; conservative row drops
- Normalization rules documented for reproducibility

REFERENCES & CREDITS

- Primary data source

- Stack Overflow Developer Survey (filtered/pre-processed dataset)

- Industry context (examples)

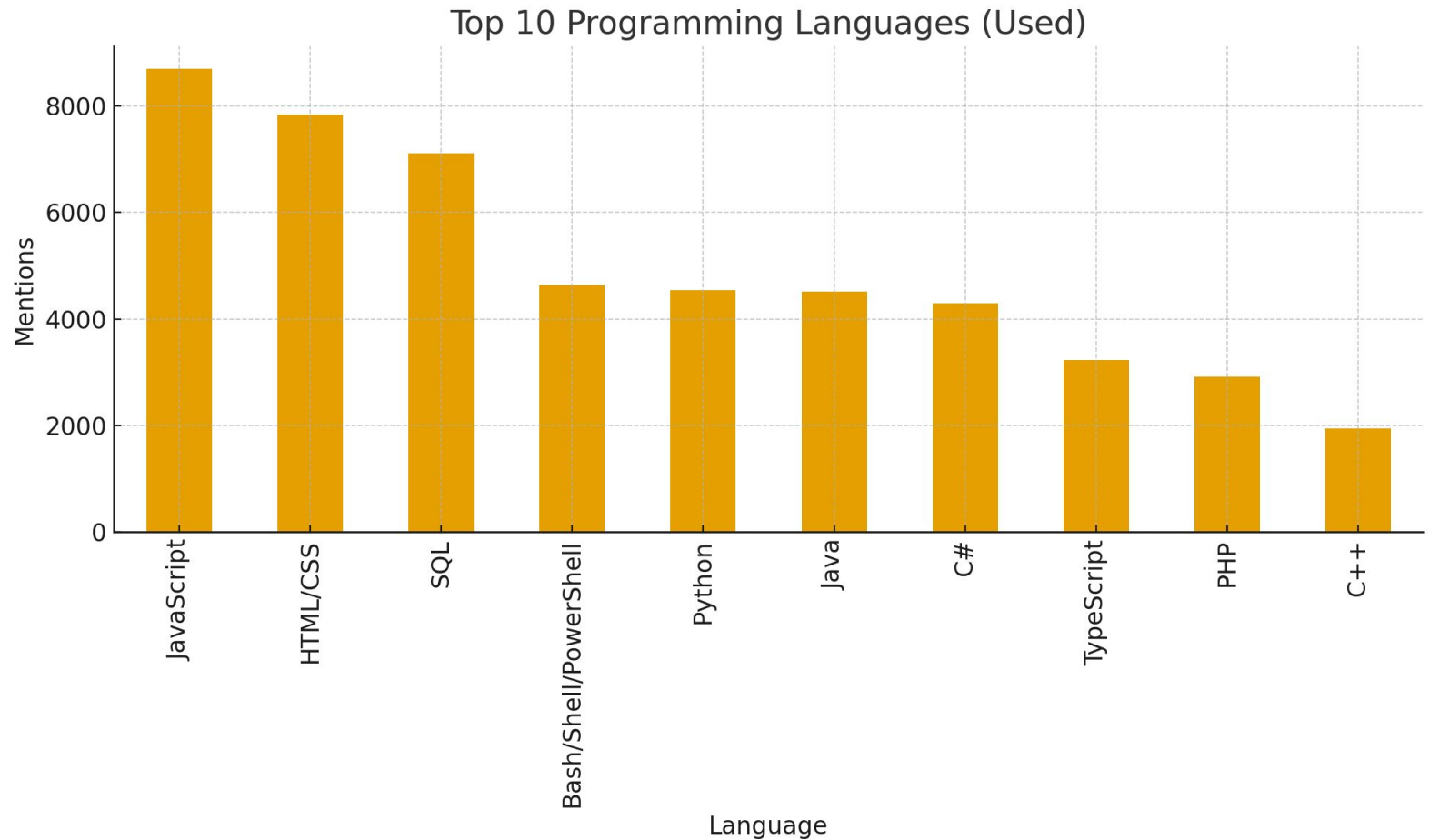
- Deloitte Tech Trends; McKinsey Technology Forward

- Tools

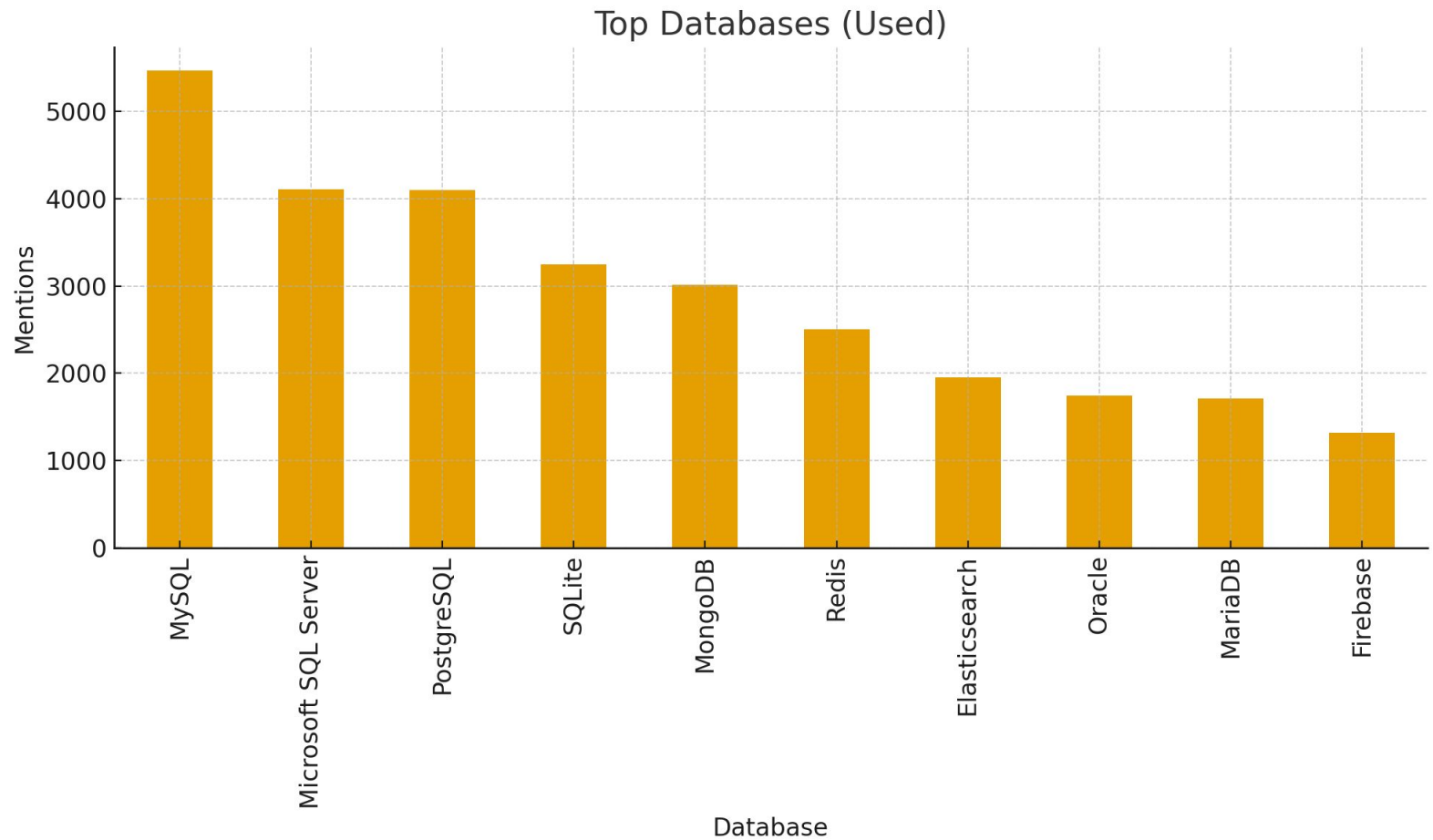
- Python (Pandas, NumPy, Matplotlib), SQL, Jupyter
- IBM Cognos Analytics, Looker Studio

- Note: Replace with formal citations as needed.

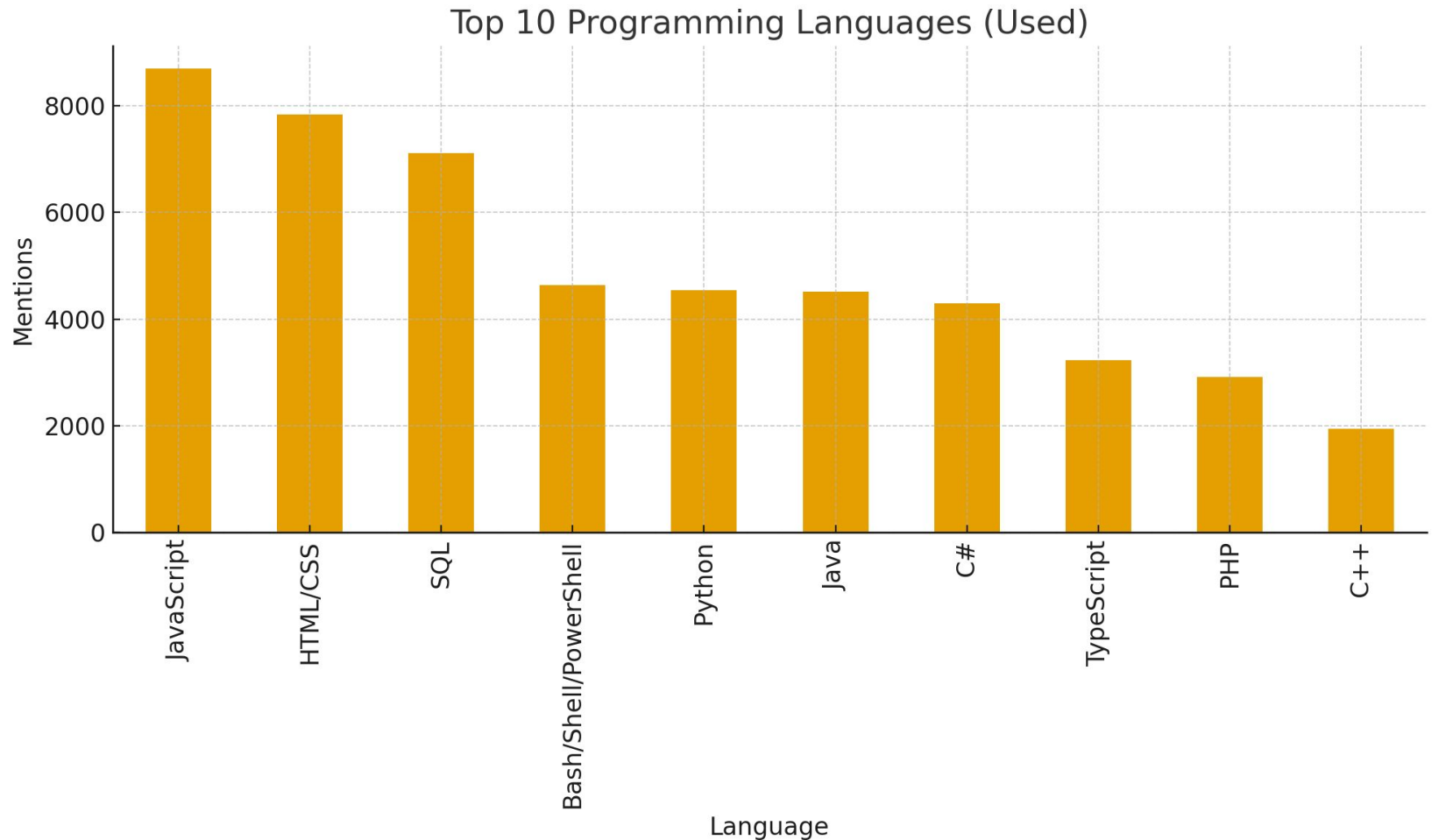
TOP 10 PROGRAMMING LANGUAGES (USED)



TOP DATABASES (USED)



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